

JunSeong Kwak

Email : junseonggg2001@gmail.com Phone : 010-9065-9248

Education

Mar.2020~ B.S., Mechanical Engineering, Double Major in AI Convergence
Feb.2026 Soongsil University, Seoul
(GPA: 4.42/4.5) graduated with highest honor (Rank: 1/123)

Research Interests

Autonomous Driving, Robotics, Model-based Control, Learning-based Control, Reinforcement Learning, End-to-End Autonomous Driving

Internships

Jan.2025~ **Mobility Intelligence and Computing System Lab (Prof. Kanghee Kim)**
Dec.2025 Physics Informed Neural Network for system identification
Model Predictive Contouring Control for autonomous racing

Publications

Dec.2025 Jimin Jeon, **Junseong Kwak**, Yunho Hong, Kanghee Kim.
Implementation and Evaluation of a Model Predictive Control Pipeline for Autonomous Racing Systems, Proceeding of the Korean Software Congress (KSC), December 2025.

Jun.2025 **Junseong Kwak**, Yunho Hong, Kanghee Kim.
System Identification and Model Evaluation of a 1:10 Scale Vehicle Using a Physics-Informed Neural Network, Proceedings of the 40th Annual Conference of the Institute of Control, Robotics and Systems (ICROS), June 2025.

Projects

Jan.2025~ **2025 The 4th F1Tenth Korea Championship (ICCAS 2025)**
Nov.2025 Developed a dual-mode control architecture integrating Model Predictive Contouring Control and Pure Pursuit for real-time autonomous racing

Jan.2024~ **2024 University Student Creative Mobility Competition (Autonomous Mobility Division)**
Nov.2024 Calibrated and fused LiDAR and Camera data for object detection, implemented a dual-mode localization system integrating LiDAR SLAM and GNSS

Jan.2023~ Oct.2023	2023 University Student Creative Mobility Competition (Autonomous Mobility Division) Implemented sensor fusion based localization using an Extended Kalman Filter by fusing GPS, IMU, and wheel encoder, implemented a vehicle lateral controller using the Stanley method for path tracking
-----------------------	--

Additional Experience

Apr.2026~ Jun.2026	H-Mobility Class, Hyundai Motor Company Studied vehicle motion control, focusing on chassis integrated modeling and longitudinal/lateral control
Sep.2025~ Dec.2025	Teaching Assistant, Autonomous Driving Programming Supported autonomous driving course projects and distributed a Docker-based race stack for student assignments
Mar.2024 Jun.2024	Autonomous Driving Programming Project Implemented and evaluated Stanley method, Pure pursuit and Stanley-with-lookahead path tracking controller in AWSIM
Jan.2023~ Nov.2024	President, Autonomous Car Control Association (ACCA) Led a student research organization focused on autonomous vehicle systems, guiding projects in hardware design, localization, control algorithms and system integration

Awards & Honors

Feb.2026	Departmental Academic Excellence Award (Mechanical Engineering), Soongsil University
Feb.2024	Top Honor Scholarship (1st Place) for SW Double Major Tutoring, Soongsil University
Sep.2023	Scholarship for Academic Excellence, Soongsil University
Apr.2021	Scholarship for Academic Excellence, Soongsil University
Sep.2020	Scholarship for Academic Excellence, Soongsil University

Skills

Programming: Python, C, C++, MATLAB
Deep Learning: PyTorch, TensorFlow
Robotics Middleware: ROS, ROS2
CAD Modeling: AutoCAD, Inventor
English Proficiency: TOEIC 835, TEPS 322